

TASK -1

House Price Prediction (Summary)

- **Objective:** Predict house prices using property features.
- **Dataset:** Housing dataset with features such as area, bedrooms, bathrooms, parking, amenities, etc.
- **About Dataset:** No missing and duplicate values.
- **Key Influencing Features:** Area, bathrooms, stories, parking, and air conditioning.
- **Model Performance:** Linear Regression performed best with an R^2 score of approximately 0.63.
- **Key Insight:** Amenities such as air conditioning and preferred location significantly affected house prices.
- **Recommendation:** Focus on larger properties with desirable amenities and prime locations to maximize property value and buyer interest.

Q1. Which features influence house price the most?

Ans: After doing correlation analysis using heatmaps it was found out that area, bathrooms, stories, parking, and air conditioning were among the most influential features affecting house prices. House area showed the strongest relationship with price.

Q2. How accurate was your model (in plain terms)?

Ans: The Linear Regression model performed reasonably well, achieving an R^2 score of about 0.63. This means it was able to capture and explain around 63% of the factors affecting house prices. While the predictions were not perfect, they were generally close to the actual prices, making the model useful for estimating property values.

Q3. What surprised you in the data?

Ans: There were a few things that make this dataset unique:

1. No missing values
2. No duplicate values
3. Also, simple features like air conditioning and being located in a preferred area had a significant impact on house prices, showing that amenities and location can be just as important as the size of the property.

Q4. One recommendation for a real estate business based on your findings

Ans: Well I think real estate businesses should really highlight properties with better amenities like air conditioners, parking lots or better locations! These features are often seen increasing the value of the properties and attracting more buyers and better sales, this reflects how people today prioritize comfort and convenience in their lifestyles, and it often starts with choosing the right home.